Credit Card Fraud Detection

By: Ryan Habis

The Dataset

Dataset contains fraudulent credit card transaction

- its been recorded in a 48 hour timeframe
- Collecting 284,807 transaction from European credit card holders.

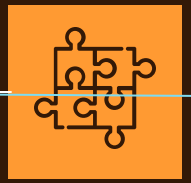
The approach used:

- CRISP-DM

What was found:

- Fraud transaction happen on smaller amount.
- Time plays a factor when fraud transaction happen.
- PCA component (V17, V14, V12) best correlated with fraud.

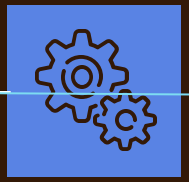
What is featured in the dataset



01

Tim
e

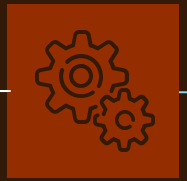
Start recording once the first transaction starts in Seconds



02

Amount

Currency is in euros



03

V1 - V28

PCA transformed for GDP

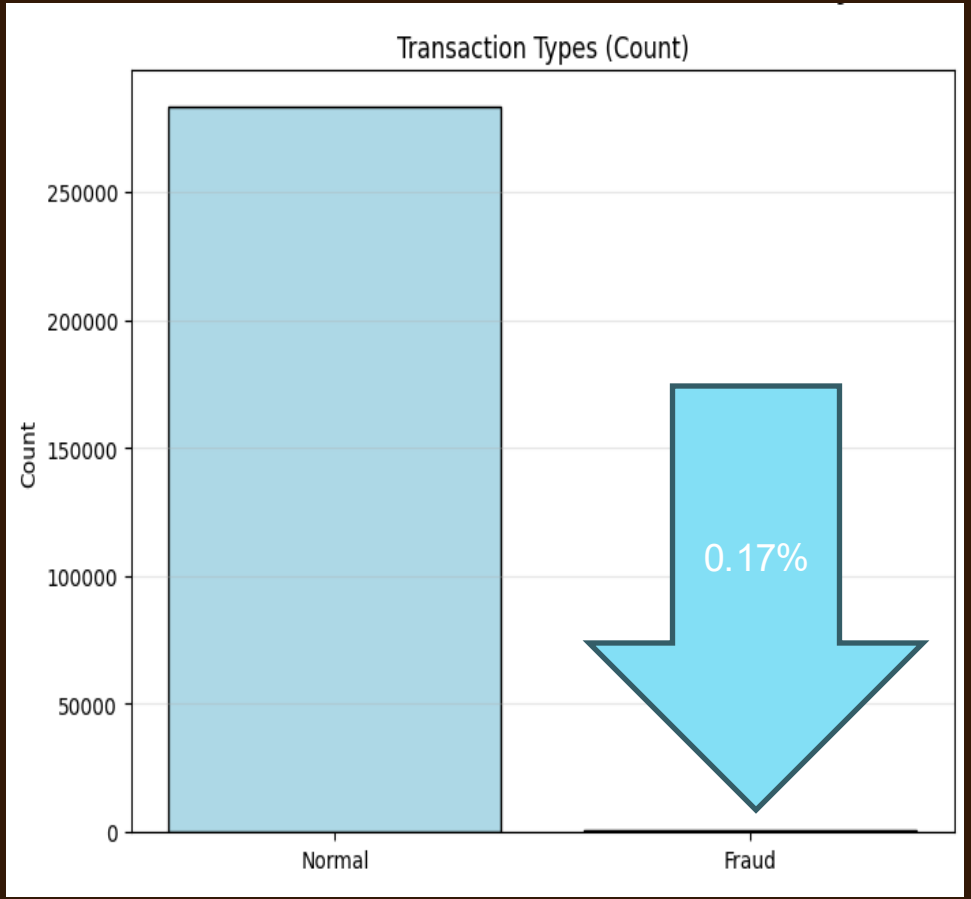


04

Class

Will show what transactions are fraud and legit

- 0.17% fraud transaction
- Ireland 160M in 2024



Research Hypotheses

Research Hypotheses

H1: Does fraud transaction amount differ from legitimate ones.

H2: Do fraudulent transaction happen more often on certain times throughout the day.

H3: Can under sampling reveal hidden fraud patterns.



Methodology

CRISP-DM

Business understanding

Fraud detection goal

Data understanding

Explore the data

Data preperation

Clean and remove dublicats and outliers

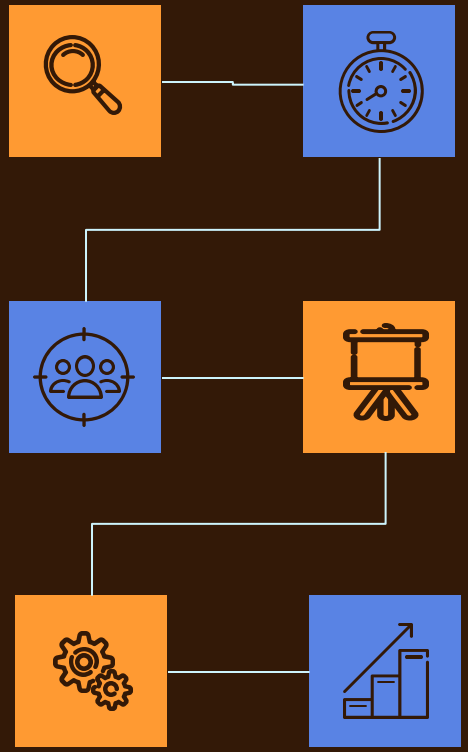
Modeling

Make a visulization

Evaluation

Test

Deployment



Analysis and Visualization

Fraud mean higher, median lower than legitimate

Bivariate Plots

Fraud started at 1 AM till 2 AM then its highest on 11 AM

Univariate Time Plots

V17,V14,V12 are the top predictors

Heatmap

V14 and V12 strongest for predicting

Logistic Regression

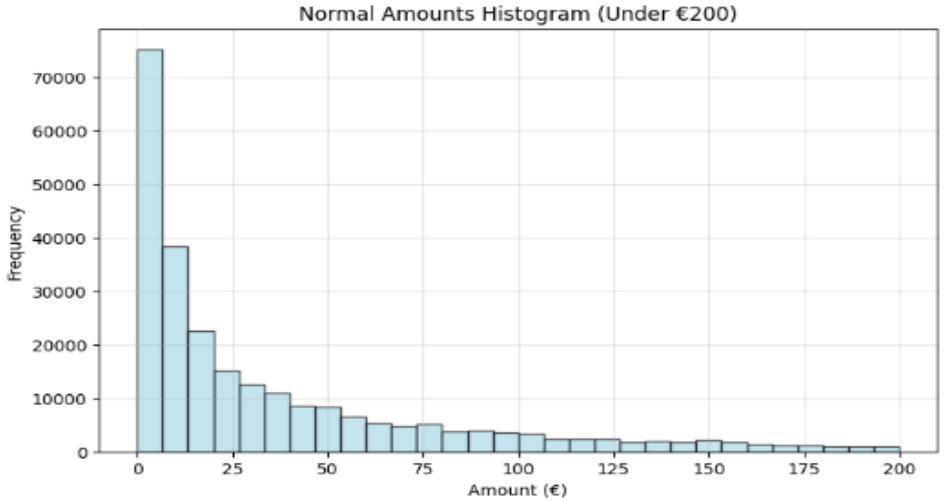
Normal mean: €88.41
Fraud mean: €123.87
Fraud mean €35.46 higher ($p < 0.01$)

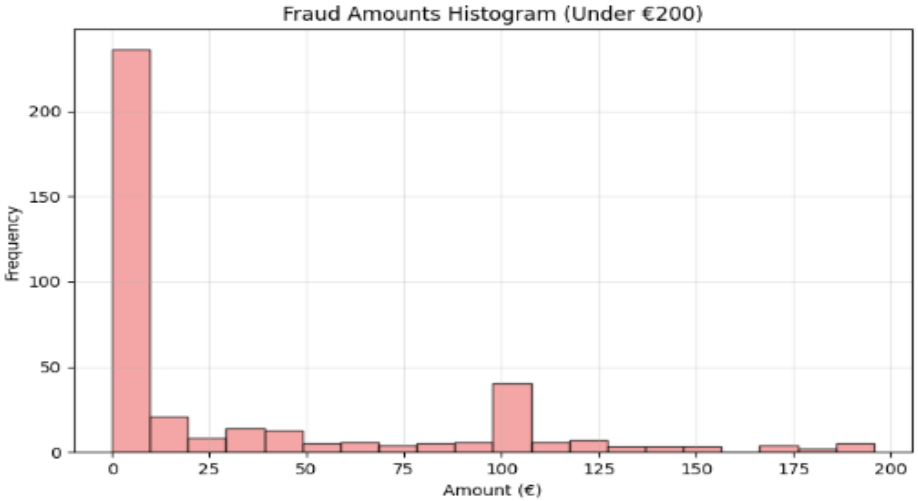
T-test

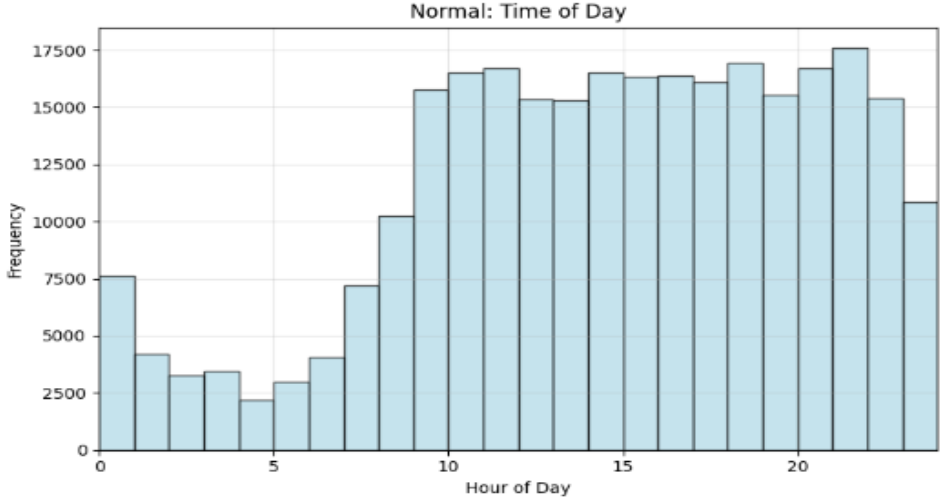
V1 differs extreamly when compaired to amount spent ($p < 0.0001$)

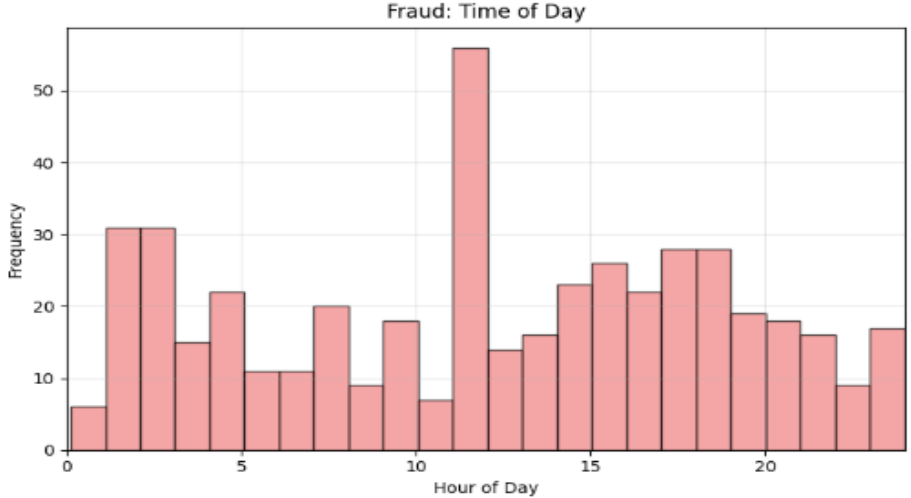
ANOVA

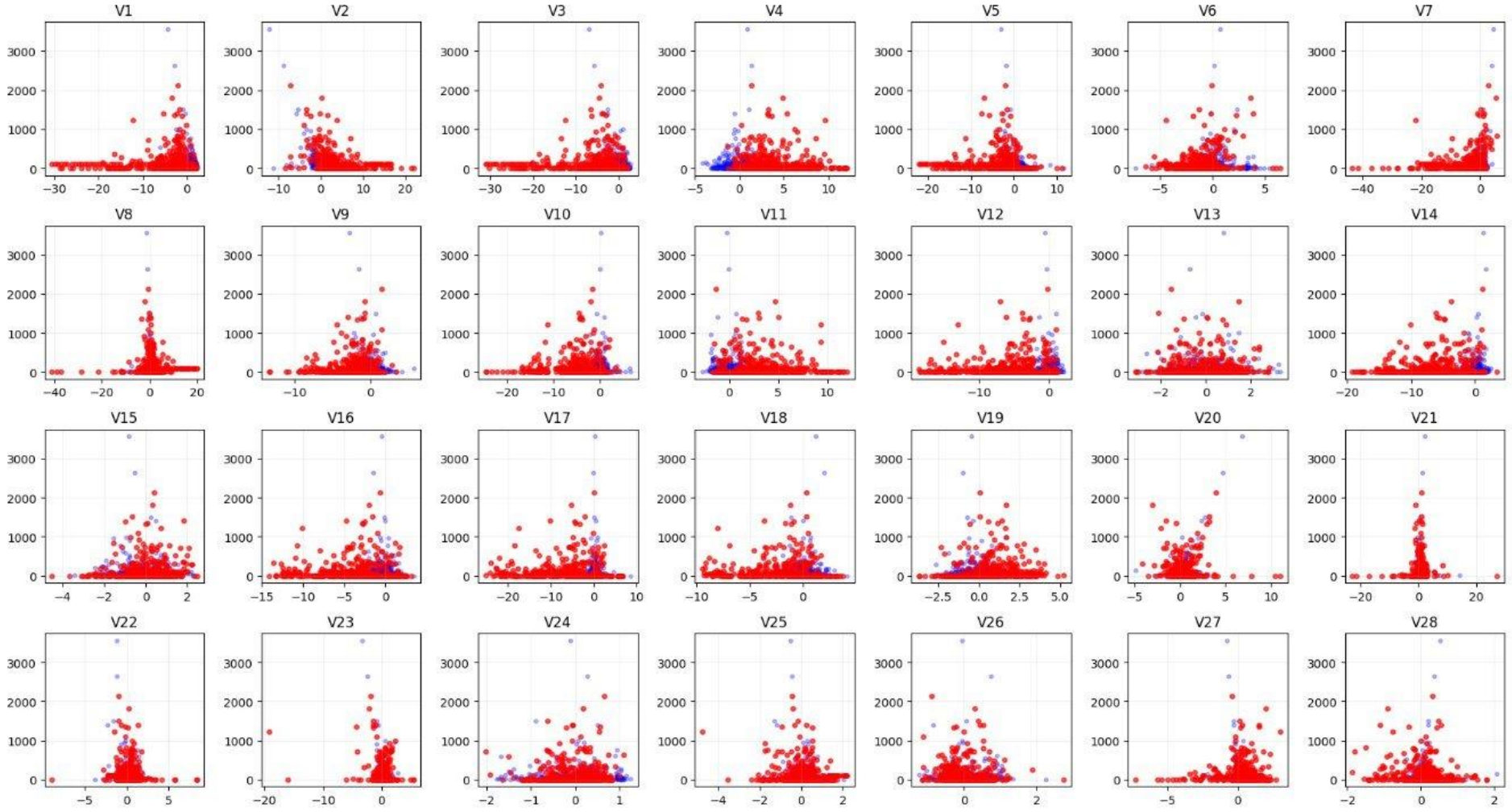
Univariate analysis: Distribution of Individual Variables







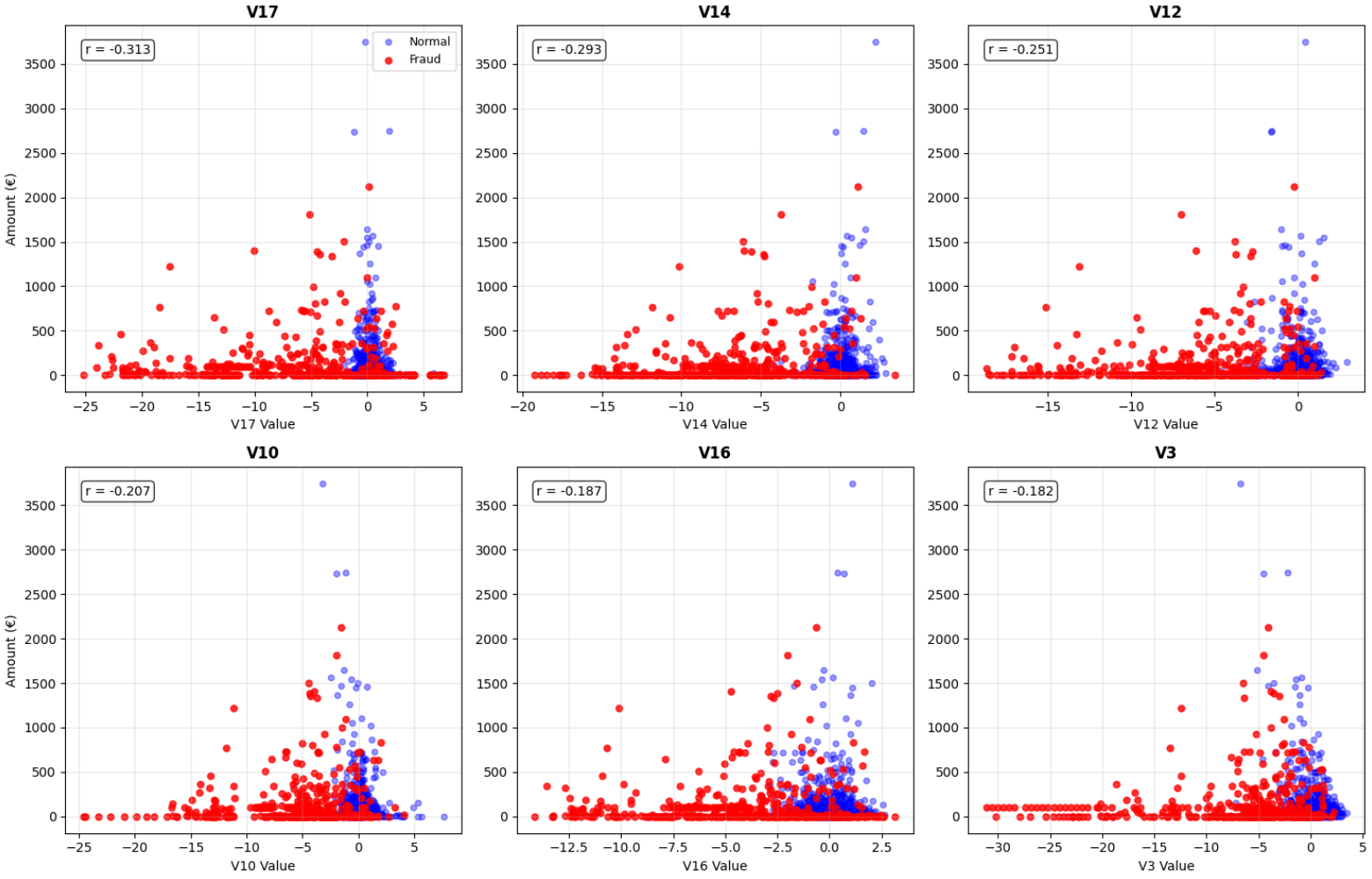



V1
V2
V3
V4
V5
V6
V7
V8
V9
V10
V11
V12
V13
V14
V15
V16
V17
V18
V19
V20
V21
V22
V23
V24
V25
V26
V27
V28

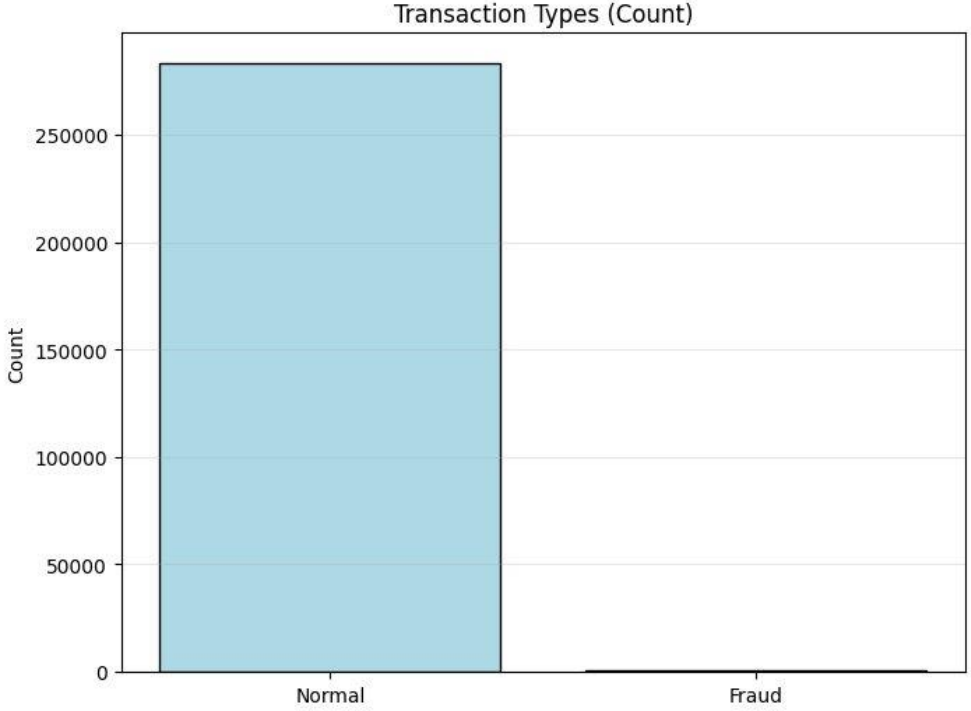
Negative coefficients the lower value means the stronger relationship to detect fraud V14 and V12 have the strongest negative coeffieients, so there the most important features in the dataset in detecting fraud

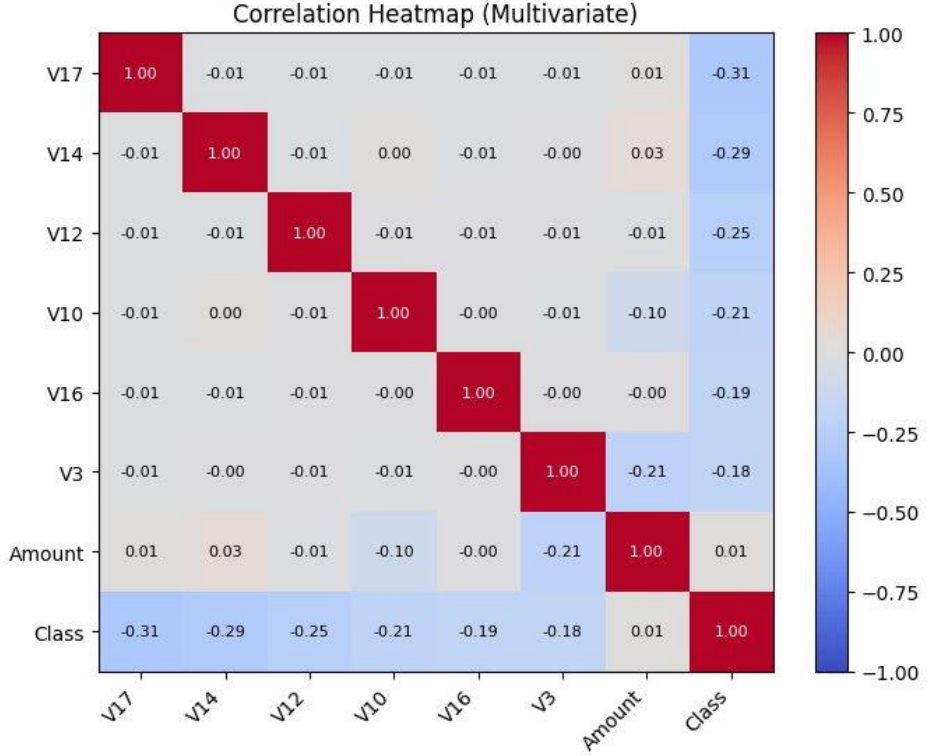
Table A.2: Logistic Regression Results for Fraud Prediction

Component	Coefficient	Interpretation
V14	-0.8647	Strongest fraud predictor
V12	-0.7565	Second strongest predictor
V16	-0.2465	Moderate predictor
V10	-0.1775	Weak predictor
V17	0.2167	Positive relationship (unusual)
Intercept	**-2.3721**	**Baseline fraud probability**


Top 6 Fraud Predictors: Relationship with Transaction Amount
V17
r = -0.313
Normal
Fraud
Amount (€)
V17 Value
V14
r = -0.293
V14 Value
V12
r = -0.251
V12 Value
V10
r = -0.207
Amount (€)
V10 Value
V16
r = -0.187
V16 Value
V3
r = -0.182
V3 Value

Multivariate analysis: Multiple Variables Together





There is a 84.3% chance that it will catch the fraud transaction, but the problem is it will also flag legitament transactions as well which leaves us with a 9% chance of getting the fraud transactions.

Table A.3: Logistic Regression Model Performance

Metric	Value	Interpretation
Overall Accuracy	0.980 (98.0%)	Very high due to class imbalance
Fraud Detection Rate (Recall)	0.843 (84.3%)	Captures majority of fraud cases
Precision	0.092	Low due to false positives
F1-Score	0.166	Reflects precision-recall tradeoff

Table A.4: Hypothesis Test H1: Transaction Amount Differences

Test	Two-Sample T-Test (Welch's)
Hypothesis (H1)	Fraudulent and legitimate transactions have different average amounts
Normal Mean	€88.41
Fraud Mean	€123.87
Mean Difference	€35.46 (Fraud higher)
T-statistic	-2.961
P-value	0.0032
Significance	**Significant at p ¡ 0.01**
Conclusion	**H1 Supported** - Fraud transactions have significantly higher average amounts

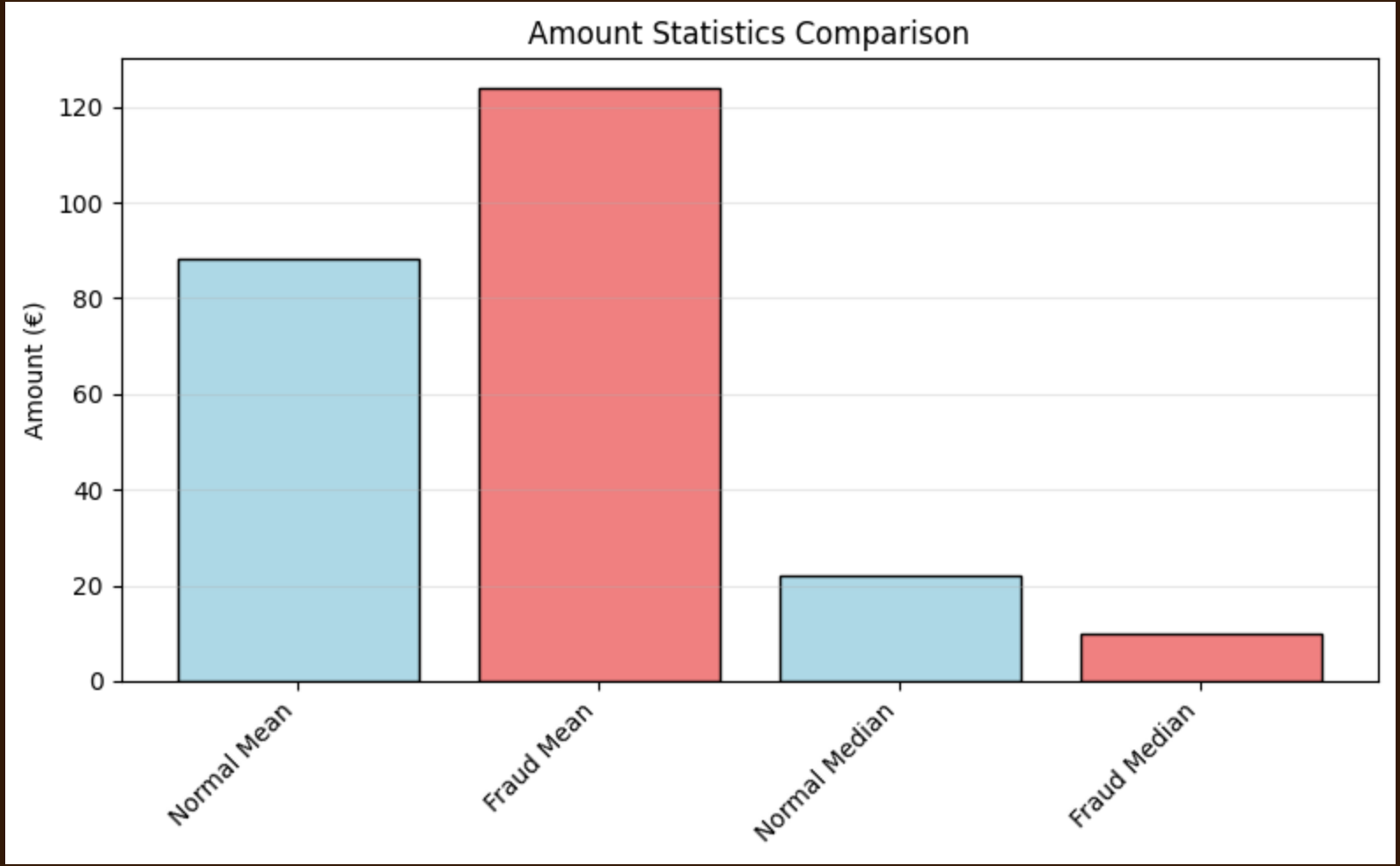

Amount Statistics Comparison
Amount (€)
0
20
40
60
80
100
120
Normal Mean
Fraud Mean
Normal Median
Fraud Median

Table A.6: Transaction Amount Categories for ANOVA

Category	Range (€)	Sample Size	Description
Small	¡ 50	192,415	Frequent, low-value transactions
Medium	50-200	69,238	Moderate everyday purchases
Large	200-1000	19,574	High-value purchases
Very Large	¿ 1000	2,499	Exceptional/luxury transactions

Table A.5: One-Way ANOVA Results: V1 Differences by Transaction Amount Category

Source	Sum of Squares	df	F-statistic	P-value
Between Groups (Amount Categories)	39,721.22	3	3630.85	¡ 0.0001
Within Groups (Residual)	1,028,040.00	281,914	-	-

Thank you

Questions?

